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Adaptive Ultrasound Image Reconstruction Based on Local Motion – Setup Script Manual

D00280 Rev. A

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Summary

This manual explains the functionality of a software-based, pixel-oriented ultrasound image reconstruction method that adapts to the motion of the media being scanned. The code integrates two different acquisition methods to optimize image quality and motion resolution simultaneously.

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Figure 4. Comparison of Data Acquisition Sequences for the AUSI approach. (left) This figure illustrates the WB spatial compounding acquisition sequence. The top panels show the 128-element apodization for receive and 32-element apodization for transmit. The bottom-left panel displays the transmit beam pattern for one of the five angles used to compound the B-mode image, and the bottom-right panel shows how the WB acquisitions are recorded, yellow rows, in a receive buffer. (right) The right side of the figure illustrates the data acquisition process for plane

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1. Overview

The fundamental idea is to overcome the traditional trade-off between image quality and frame rate in ultrasound imaging. Conventional methods apply a single acquisition scheme to the entire image, which limits high-quality imaging to situations with minimal motion. This code addresses this limitation by adapting the image reconstruction at each individual pixel based on the local motion detected at that point.

- **Low Motion Regions**: In areas with slow or no motion, the code uses longer acquisition periods and combines more data to create a higher-quality image with better spatial and contrast resolution.
- **High Motion Regions**: In areas with fast motion, the code uses shorter acquisition periods and fewer data points to ensure that motion details are captured accurately.

This approach produces a final, composite image where different parts are reconstructed using varying amounts of data, leading to an overall improvement in image quality without sacrificing motion resolution.

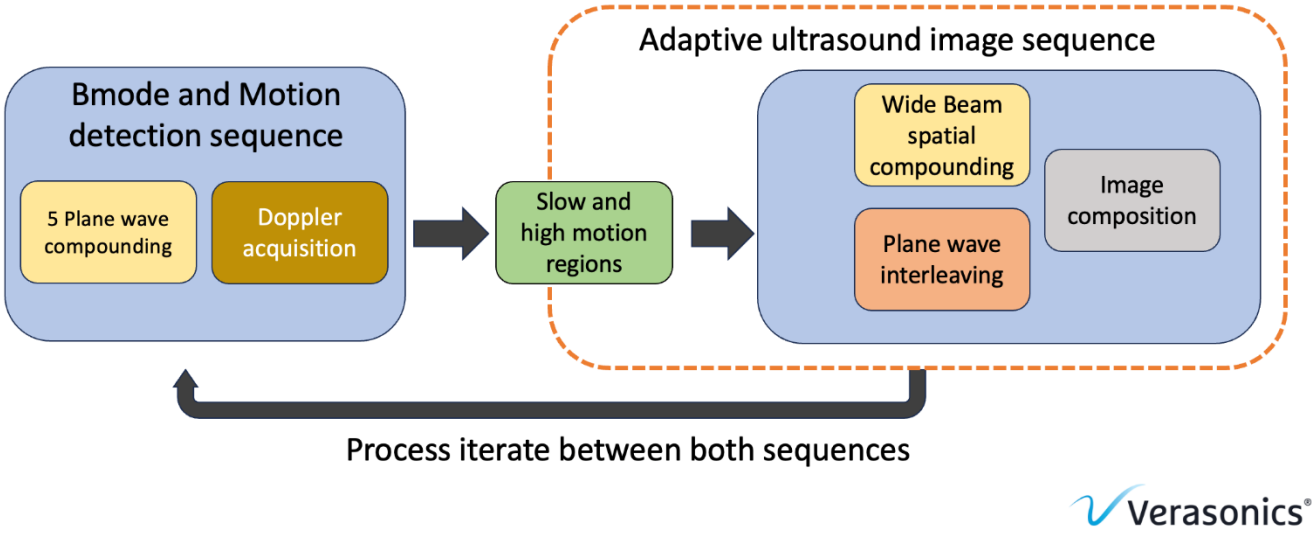

Figure 1. Schematic illustration of the Adaptive Ultrasound Imaging sequence using a Verasonics system.

2. Acquisition and Motion Detection

The system operates by interleaving two types of acquisition sequences to perform both real-time motion detection and image acquisition for reconstruction.

2.1 Initial B-mode and Motion Detection Sequence

This sequence is primarily for detecting real-time spatial motion using a simple Doppler-based approach. The user could replace this method for any other it may find for better speckle tracking and motion detection. It also generates an initial B-mode image.

2.1.1 Plane Wave Compounding

The sequence begins with plane wave plane wave acquisitions, using 5 different steering angles. This method is used for an initial B-mode image and has a total pulse repetition frequency (PRF) of 1.25 kHz.

2.1.2 Fast Media Tracking

For motion detection, the system performs a sequence of 20 transmit/receive (Tx/Rx) cycles per ensemble, at a PRF of 3 kHz.

For both acquisitions, a full 128-channel apodization is used.

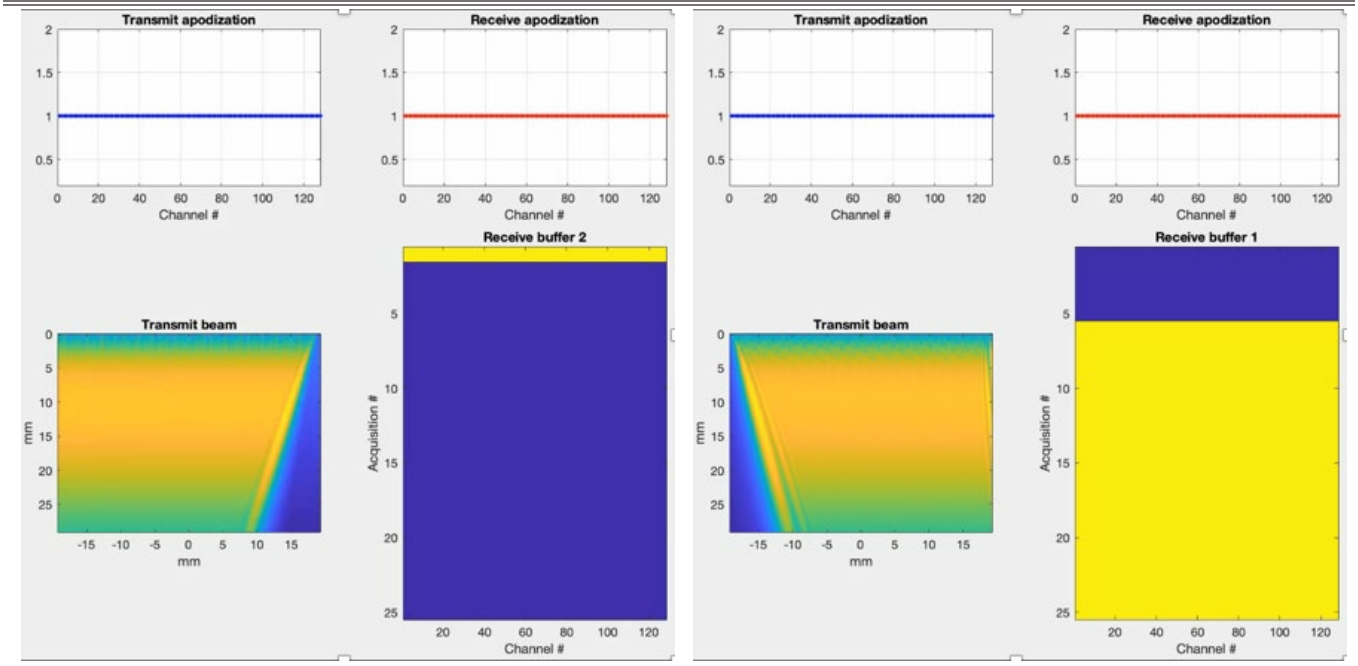

Figure 2. Comparison of Data Acquisition Sequences. (left) This figure illustrates the plane wave compounding acquisition sequence. The top panels show the 128-element apodization for both transmit and receive. The bottom-left panel displays the transmit beam pattern for one of the five angles used to compound the B-mode image, and the bottom-right panel shows how the first of five acquisitions are recorded in a receive buffer. (right) The right side of the figure illustrates the data acquisition process for Doppler-based motion detection. The top panels again show the 128-element apodization for transmit and receive. The bottom-left panel shows the transmit beam pattern of a single transmit beam, while the bottom-right shows how the last twenty acquisitions are stored in a receive buffer for processing.

2.1.3 Motion Classification

Using data from motion tracking, the code analyzes speckle patterns to estimate the motion at each pixel. Based on a defined threshold (defined as 'pwrThres' in the script), a Doppler method classifies these pixels into "fast motion" and "slow-motion" regions.

The external function **GetRegion**, located in the **Scripts** folder, then assigns the corresponding pixels for each region to the PData structure. It takes the second image buffer as its input. Users can also use their own external functions to calculate media motion and provide that data to the second image buffer. **GetRegion** will then use this output to update PData(1) with a new PData(1).Region structure that identifies both the fast and slow-motion regions.

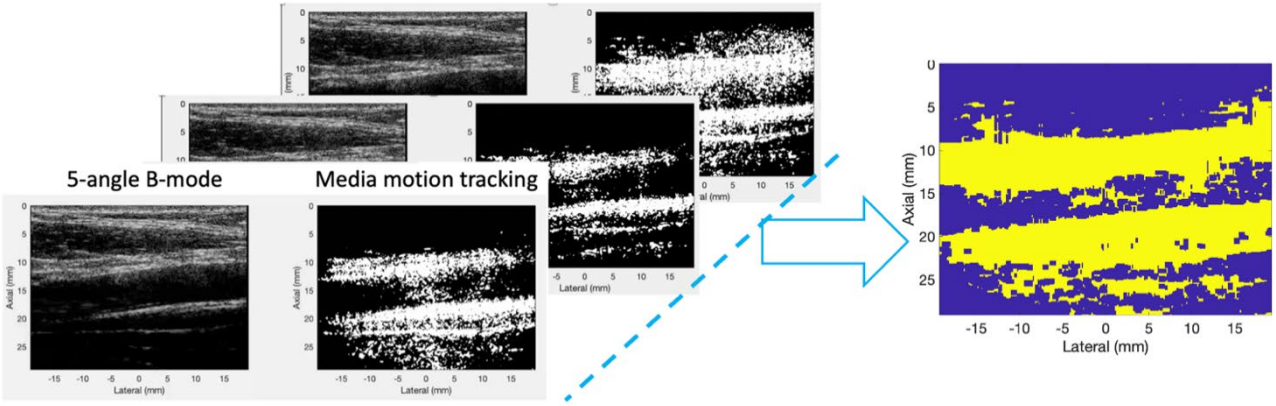

Figure 3. This figure illustrates the workflow for real-time motion detection and classification during a B-mode ultrasound scan. As the script begins, it displays the initial B-mode image and, in parallel, analyzes the scanning area to detect significant movement. Based on a predefined threshold, the script successfully detects and classifies regions with faster motion, such as the carotid wall and blood flow. The detected motion frames are saved to a ***second image buffer****. These frames are then processed by the external function* ***GetRegion****, which averages them to create a final classification map. The function assigns the corresponding pixels to 'slow' and 'fast' motion regions within the* ***PData.Region*** *structure. The final output, represented by the blue/yellow image, shows these classified regions for analysis.*

3. Acquisition and Adaptive reconstruction

This is the main reconstruction sequence, where different methods are applied based on the motion classification.

3.1 Adaptive Ultrasound Image (AUSI) Sequence

3.1.1 Interleaved Wide Beams Acquisition

An interleaved Wide Beam (WB) transmit/receive event is performed using 48 wide-focused beam acquisitions. These acquisitions are performed using spatial compounding with 5 different steering directions. The transmit aperture size is set to 32 elements, while the receive aperture uses all 128 elements.

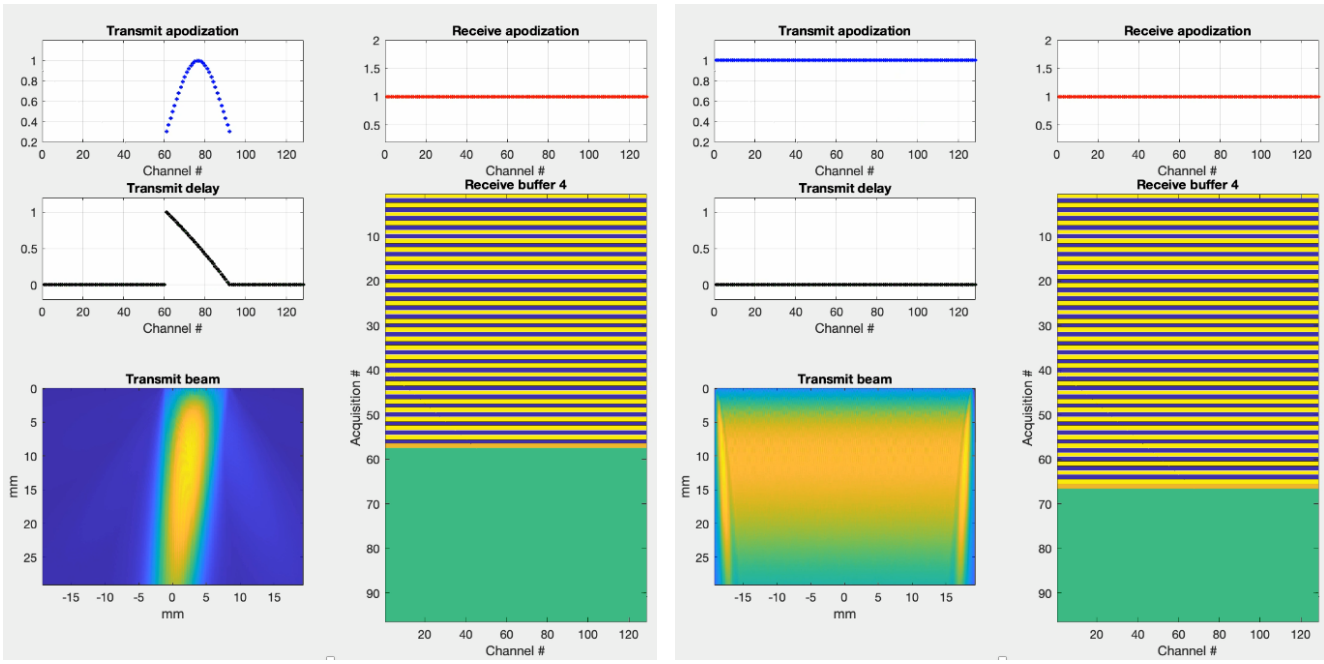

Figure 4. Comparison of Data Acquisition Sequences for the AUSI approach. (left) This figure illustrates the WB spatial compounding acquisition sequence. The top panels show the 128-element apodization for receive and 32-element apodization for transmit. The bottom-left panel displays the transmit beam pattern for one of the five angles used to compound the B-mode image, and the bottom-right panel shows how the WB acquisitions are recorded, yellow rows, in a receive buffer. (right) The right side of the figure illustrates the data acquisition process for plane wave transmits. The top panels again show the 128-element apodization for transmit and receive. The bottom-left panel shows the transmit beam pattern of a single transmit beam, while the bottom-right shows how the plane wave transmit data, blue color, is stored in the same receive buffer for processing.

3.1.2 Interleaved Plane Wave Acquisition

An interleaved Plane Wave (PW) transmit/receive acquisition is performed concurrently with the wide beam scanning. This data is exclusively used for image reconstruction in the fast-motion regions. All 128 elements are active for transmit and receive.

Table 1. Interval and frequency repetition parameters used in the AUSI script

	Pulse Repetition Interval	**Pulse Repetition Frequency**
Each acquisition	160 µs	6.25 kHz
Plane wave frame	320 µs	3.13 kHz
Wide Beam frame	320*48 = 15.36 ms	65.10 Hz

3.1.3 Slow and Fast-Motion Region Reconstruction

In regions classified as "slow-motion," the code reconstructs the image by accumulating IQ (In-phase and Quadrature) data from 48 wide-focused beam acquisitions. The frames are then averaged using a running average that updates with each new frame. In regions classified as "fast motion," the code uses flash plane wave acquisitions for reconstruction. This method uses a single transmit pulse to generate the entire image, which is ideal for capturing fast motion.

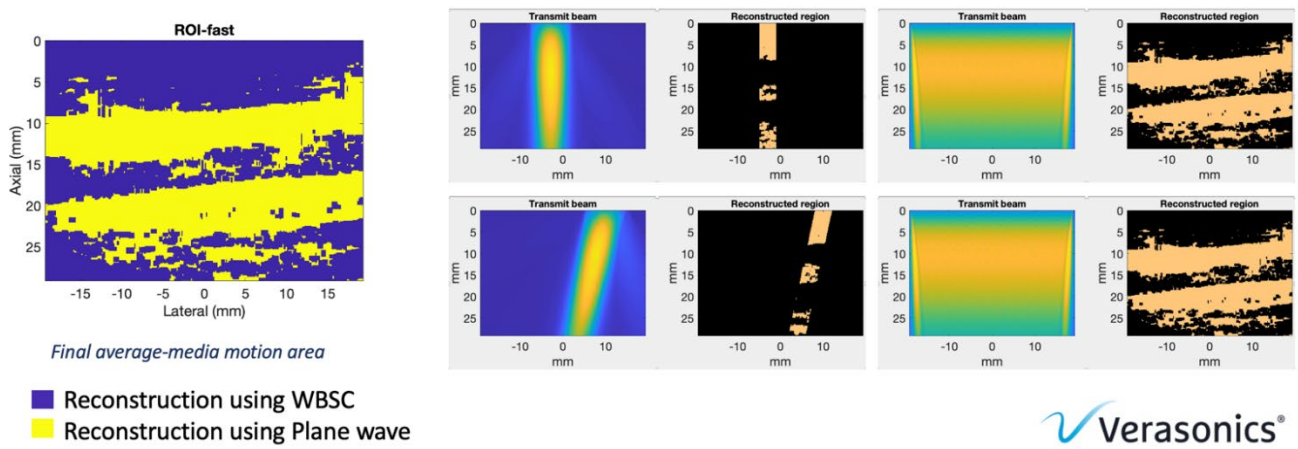

Figure 5. This figure compares the pixel areas reconstructed using different data acquisition methods. On the left, the blue/yellow image shows the final classified regions; blue pixels were reconstructed using WB transmits, while yellow pixels were created with PW transmits. The middle panel illustrates two different wide beam patterns, with the gold-colored areas showing the specific pixel areas reconstructed by each WB acquisition. The right panel shows the pixel areas reconstructed by the plane wave transmits, with the gold areas corresponding directly to the yellow regions in the classified image on the left.

3.1.4 Final adaptive image composition

Once the software reconstructs the data for the slow and fast-motion regions, it sends the reconstructed data to image buffers 3 and 4, respectively.

The Verasonics 'Image' process then combines this data in two steps:

1. It uses image buffer 3 and the 'lowerHalf' mapping method to display the slow-motion region.
2. It continuously updates the fast-motion region by calling the Image process again, this time using image buffer 4 and the 'upperHalf' mapping method.

The final adaptive image is a composite of these two processes, combining the slow-motion data with a continuously updated view of the fast-motion data.

4. Summary of Operations

The entire process works as an adaptive loop:

1. Motion Detection: The plane wave sequence quickly estimates media motion at all points in the image.
2. Pixel Classification: Pixels are classified as fast-motion or slow-motion based on a Doppler threshold.
3. Adaptive Reconstruction:
    - Slow-motion pixels: Reconstructed using the high-quality, wide-beam spatial compounding data.
    - Fast-motion pixels: Reconstructed using the high frame rate, flash plane wave acquisitions.
4. Final Image: The system composites the different reconstructed regions into a single final image, with each area optimized for its specific motion characteristics.

5. Running the script

After you 'activate' your Verasonics release software, run the AdaptiveUltrasoundImaging_script.m script

```
>> AdaptiveUltrasoundImaging_script
>> VSX
Name of .mat file to process: AdaptiveUltrasoundImaging_script
```

After you execute the script using VSX, two windows will appear: the initial B-mode image and the VSX Control GUI. Follow these steps:

- Adjust the transmit voltage for the B-mode image using the High Voltage P1 slider.
- Check and adjust the TGC sliders for your specific scanning needs.
- Increase the transmit voltage for the Doppler transmits using the High Voltage P2 slider.
- Adjust the TGC for the Doppler signal. You can do this by using the small box located next to the TGC title in the upper-left corner of the GUI.
- The Doppler power threshold slider (DopPwThres) controls the Doppler sensitivity. Adjusting this slider or its value will modify the number of pixels assigned to the "fast motion" region.
- Click the AUSI button to start the Adaptive Ultrasound Image (AUSI) sequence. A new window will appear, displaying the slow and fast-motion regions in blue and yellow, respectively. After a few seconds, the AUSI image will be displayed in the main image window.
- Click the Get_media button to return to the initial sequence if you want to:
    - Recalculate the slow and fast regions using a new threshold.
    - Move to another region of interest to scan.
- When you close the GUI, you can access the raw RF data in the RcvData structure and the image data in the ImgData structure. Specifically, ImgData{3,1} and ImgData{4,1} save the reconstructed images of the slow and fast-motion regions, respectively. Additionally, a fifth image buffer has been created to save the composite AUSI images.

An alternative script, AdaptiveUltrasoundImaging_wLogo_script.m, is provided for users who wish to define a fixed, arbitrary region instead of using the region calculated by the Doppler processing.

This script utilizes the createVerasonicsV function, which takes PData(1).Size(1) and PData(1).Size(2) as inputs and outputs a vector to replace the last PData(1).Region.PixelsLA variable. This parameter then defines the 'V' shape for the fast-motion region. Users can replace this function with their own to create any custom, fixed region for the Adaptive ultrasound imaging sequence.

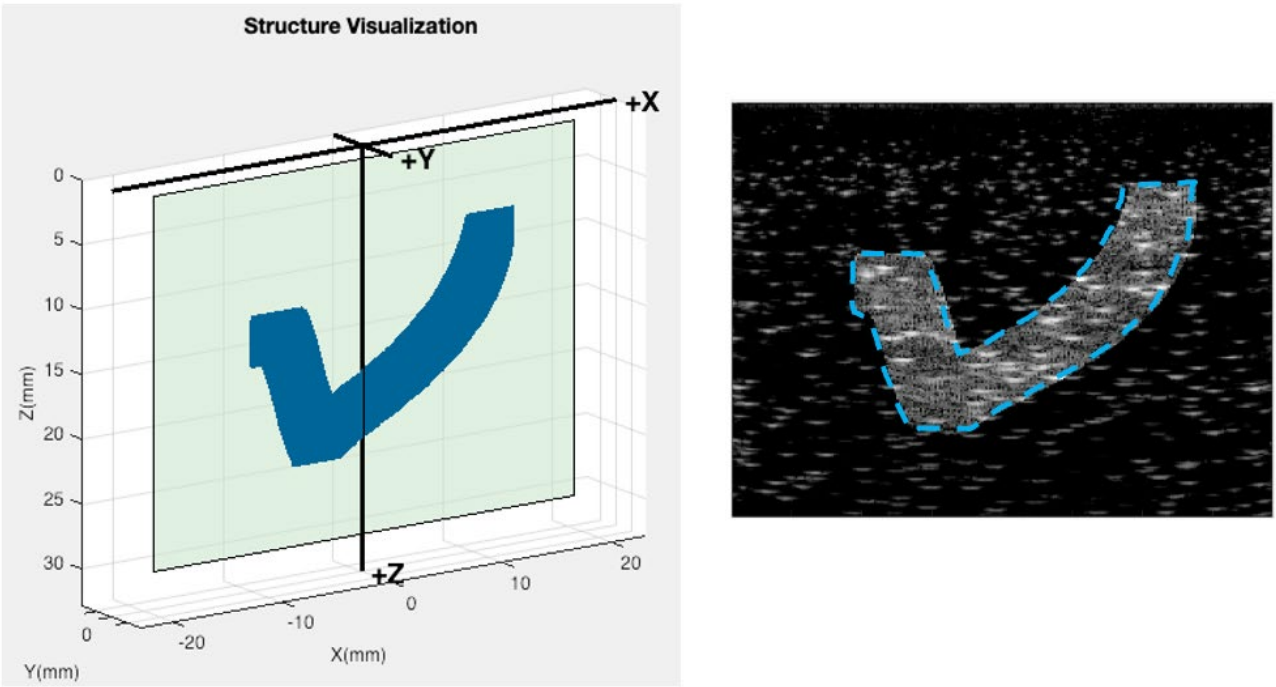

Figure 6. This figure illustrates the use of an arbitrary shape. The left panel shows a 'V'-shaped region, representing a user-defined area for the Adaptive imaging. This region, inspired by the Verasonics logo, is created for a fast frame rate reconstruction. The right panel shows a microbubble flow field, demonstrating the effectiveness of this approach. Inside the 'V' shape, microbubbles exhibit a continuous flow movement due to the higher frame rate, while outside this region, their movement appears discontinuous. The fast-motion region (inside the 'V') uses plane wave transmits, resulting in lower contrast and artifacts from side lobes, but the slow-motion region (outside the 'V') maintains higher image quality, contrast, and spatial resolution.

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